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RESEARCH ARTICLE



Building and exploiting a Digital Twin for the management of drinking water distribution networks

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ABSTRACT

Digital Twins (DTs) are starting to be exploited to improve the management of water distribution systems (WDSs) and, in the future, they will be crucial for decision making. In this paper, the authors propose several requirements that a DT of a water distribution system should accomplish. Developing a DT is a challenge, and a continuous process of adjustments and learning is required. Due to the advantages of having a DT of the WDS always available, during the last years a strategy to build and maintain a DT of the water distribution network of Valencia (Spain) and its Metropolitan Area (1.6 million inhabitants) was developed. This is one of the first DTs built of a water utility, being currently in operation. The great benefits of their use in the daily operation of the system ensure that they will begin to be usual in the most advanced smart cities.

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KEYWORDS

Water Distribution Systems; real time; detailed model; Digital Twin; planning operation and maintenance; Epanet

Introduction

There is a growing need for more sustainable and efficient cities nowadays, due to their increasing population and high resource consumption (Mohammadi and Taylor 2018). In that context, informed decision making should employ advanced tools that combine real-time sensor data gathering, advanced analytics and model-based capabilities to simulate ‘what-if’ scenarios. As defined by various authors, a ‘Digital’ or ‘Virtual’ Twin of a defined system would fulfil those requirements.

The Digital Twin philosophy has been used since the 1960s by the National Aeronautics and Space Administration (NASA) because of the need to remotely operate and maintain the systems, although the Twin was then physical rather than virtual. As an example of its usefulness, the NASA could successfully perform the rescue mission of the Apollo XIII thanks to its twin on earth.

The concept of Digital Twin (DT) was definitively consolidated by M. Grieves in 2003 (Grieves 2015). It was first aimed at optimizing the life cycle of a product, in relation to its design and manufacture process, and with the subsequent maintenance during its lifetime. According to Grieves, a Virtual Twin model must contain three main parts: the physical assets, a virtual model, and the connections of data and information that tie the virtual and real spaces together.

Since then, numerous researchers have defined in more detail the theoretical foundations of DTs applied to the industry field (Qi and Tao 2018; Tao et al. 2018, 2019; Tao and Qi 2019). Most of the works highlight the following key technologies involved in building a DT:

- *Modelling*: physical and virtual models must describe the main features of the system.

- *Connection*: the physical and virtual system should be constantly connected. This concept includes data transmission, conversion, storing, protection, etc.
- *Data advanced analytics*: in order to get information from a dataset, it must be pre-processed (cleaned and filtered) and mined by data analysis techniques and Artificial Intelligence (AI) algorithms.
- *Interaction and service*: the DT should be able to suggest, operational settings, once validated by simulations, to optimize or to adapt the system processes to external changes.

DTs are not only useful in industry, but they can also be developed and exploited in a city management context, particularly in the drinking water supply system. Hydraulic models are more and more necessary and commonly exploited for decision-making by Water Utilities (WU). Water Distribution Systems (WDSs) have become complex hard-to-manage systems, since WU must deal with challenges like the population growth, the resource scarcity or the infrastructure wear and tear. Therefore, WDSs are also on the scope of the new technologies (Sarni et al. 2019), and the Digital Twin concept is starting to be used in order to solve many of the problems that concern the management of those systems (Shi, Xu, and Du 2019), like asset management, leak localization, optimization of the system operation, energy efficiency, water quality, planning of maintenance operations, early response to emergencies, etc. In this way, the DT will be an essential support system in the near future for decision-making (Chacón, Albarrán, and Cruz 2019).

There are few examples of precursors of a DT in WDSs. Lin, Sedigh, and Miller (2009) proposed a preliminary Cyber-Physical System (CPS) combining the Epanet hydraulic simulator (Rossman

2000) with the mathematical Matlab software (Pratap 2002) to build an intelligent cyber WDS that can operate certain valves to optimally allocate demands with the goal of distributing the water as equitably as possible. The authors present it as a basic example of an intelligent decision support system.

Wang et al. (2015) focused their work in the definition of a CPS for the management of the water quality main parameters. According to Wang et al., the CPS needs: a) sensors and an effective communication network, b) computing technologies to build models, manage the data and perform data analysis, and c) predictive control technologies.

The United States Environmental Protection Agency (EPA) and the University of Cincinnati created the Epanet-RTX extension, a real-time hydraulic modelling framework based on the Epanet engine (Hatchett et al. 2011). The application, which permits the connection of a hydraulic model with real-time field data, was created with the aim of helping to understand the hydraulic and water quality behaviour of WDSs and explore the potential of hydraulic models to represent the real systems. Its capabilities were successfully tested at field scale for the case study of Northern Kentucky Water District (Uber et al. 2014).

Kartakis, Abraham, and McCann (2015) built a small-scale test bed, which could be viewed as a physical twin able to simulate a real WDS, considering multiple failure and anomaly scenarios, like pipe bursts or communication issues. Their approach allowed running real-time control algorithms fed from sensors data and aimed at energy optimization, automatic control and event-driven control.

The aforementioned studies share common attributes with DTs as defined in the industrial sector, however, examples of the application of a DT in the operation and management of a real WDS are scarce and generalist. The main novelty of this work is to present how the concepts and philosophy of DTs have been successfully applied to the management of a real WDS.

Objectives

One of the objectives of the present work is to propose a detailed description of the structure and the set of features that define a WDS Digital Twin, which must replicate the real system behaviour in a virtual model that serves as the basis for experimentation, i.e. a virtual copy of reality that allows to perform any simulation. In fact, for WDSs, a complete DT would be a virtual replica of all the processes taking place from the raw water catchment until the users tap.

In this way, a DT would allow monitoring the whole water cycle. The DT should analyse the performance of every phase by simulating the system behaviour under any scenario and request. It would also permit to improve the system performance and efficiency thanks to the ability to make informed and optimized decisions and, therefore, reduce risks.

However, this work is focused on the development and use of a DT just for the water distribution stage, where it is used in order to solve many of the problems concerning the distribution network management, from the design to the daily operation. The main capabilities of the proposed DT are the following:

- Optimal design of the network elements, such as pipes, valves, tanks or pumping stations, with the goal of minimizing the overall carbon footprint and subject to quality of service constraints, taking into account the demand forecast and the expected future extension of the network.
- Master planning, considering the extension of the network, the replacement strategy and the requirements to improve the system resilience. The DT will also define the strategic operations to be done when integrating the new assets in the system.
- Model based leak detection and pre-location, which will significantly reduce the leaks lifetime. Hence, the DT will help to preserve such a scarce resource, as well as to reduce operational costs.
- Determination of the optimal daily operation parameters with multiple objectives. These will include water velocity in pipes, service pressure and energy efficiency, among others.
- Early warning and informed response to emergencies, which will help the operators to make quick and effective decisions.
- Water quality prediction, which combined to online measurements of the main parameters will allow to detect quality weaknesses throughout the network and to find the best strategy for their improvement.

The next sections describe in more detail about the requirements and capabilities that a DT must satisfy in order to accurately replicate the WDN behaviour, some practical issues about its implementation and a real use case of DT in the WDN of Valencia (Spain) and its Metropolitan Area.

Materials and methods

WDS Digital Twin requirements

As highlighted above, although there are many publications about the advantages of DTs in general, few of them concern WDSs. In fact, there is not a precise definition of the features that a DT of a WDS should have in order to differentiate it from other similar applications. From the authors point of view, a true DT of a WDS, according to the given definition and in order to be useful, must meet the following requirements:

- It must account with absolute accuracy and fidelity for all the elements that are part of the distribution network, and that can affect water flow regimes, water quality and maintenance operations.
- It must accurately reproduce the behaviour of all the network elements, even in extreme operating conditions.
- The DT should also be able to exactly simulate all the controls programmed in remote terminal units (RTUs), including the emergency response sequences.
- Also, it must be able to reproduce the network behaviour during all the maintenance operations including, for instance, tank cleaning operations by considering the

different containers and auxiliary pipes to account for the storage capacity in any scenario, valve closure due to repairs, pipe filling and emptying conditions, etc.

- It must reproduce the demands on the service connections as accurately as possible. In this way, the consumptions should be registered with the highest available temporal and spatial resolution, being the last aim to register all hourly-metered consumptions without delay.
- The DT should mimic the observed real behaviour of the network as accurately as possible, according to the data registered by all available sensors, concerning levels, pressures, flows, water quality, etc. With this purpose, the model must be perfectly calibrated and must be able to reproduce all the control operations performed on the network, either manual or programmed.
- The last important aspect to be highlighted concerns to the reliability of the measurements collected by the sensors, which will be used to assess the degree of adjustment of the twin to reality. Reliability is also fundamental in real time decision making, and especially in alarm detection. A powerful anomaly detector must be developed, provided with tools to correct faulty or wrong values, based on the historic records and patterns, in order to feed properly the DT.

Nowadays it is possible to develop a DT because there is a lot of digital information allocated in different systems. Internet of Things (IoT) technologies have allowed the installation of sensors in isolated areas, where previously infeasible, thanks to the low energy consumption of the ultra-narrow band (UNB) based IoT devices. But UNB is only one of the communication technologies available today. Among the set of Information and Communication Technologies (ICTs), data can be retrieved by multiple means, including wired or wireless communications, such as internet, radio, etc. However, this information, in order to be useful, must be filtered, normalized and properly integrated, so a big data platform is necessary to develop a DT.

The main data sources managed by Digital Twins of WDSs are Geographical Information Systems (GIS), Automated Meter Readings (AMR), Computerized Maintenance Management Systems (CMMS), and field data stored by Supervisory Control And Data Acquisition (SCADA) systems. In order to keep the DT always ready to use, it is important to be aware on the frequency of change of the data from the different sources. For instance, information from GIS should be updated every week at least, or when substantial changes in the network take place; AMR should desirably be available on an hourly basis, CMMS data must be considered when the operations take place, while SCADA data should be updated every 5 minutes, or even every minute when operating in real time.

Also, in order to be able to simulate any scenario, the DT must work upon a hydraulic model that provides coherent and accurate results of the simulation. The hydraulic model must be accurately developed and calibrated, and as a part of the DT, it must be live and constantly paired to the physical world to which it represents, so the hydraulic model must be fed in a continuous way from all the information sources.

Finally, the DT not only has to be able to simulate hypothetical scenarios, but it also must include data analysis and optimization tools. Machine Learning algorithms, such as decision trees, support vector machines or artificial neural networks, can provide the capability to model complex sub-systems not included in the hydraulic mathematical model, particularly they can be used for demand forecasting or data filtering. Moreover, the DT can include optimization algorithms, i.e. linear and non-linear programming, metaheuristics, etc., to generate optimal operation rules or optimal design based on different criteria (energy, water quality, quality of service, etc.).

Therefore, integrating the hydraulic model with all the information sources, and using advanced analytics, a DT is obtained and all its potential can be exploited (Figure 1).

WDS Digital Twin implementation

Getting to these levels of detail is often a challenge, so the DT is not developed overnight. It consists of a continuous process of

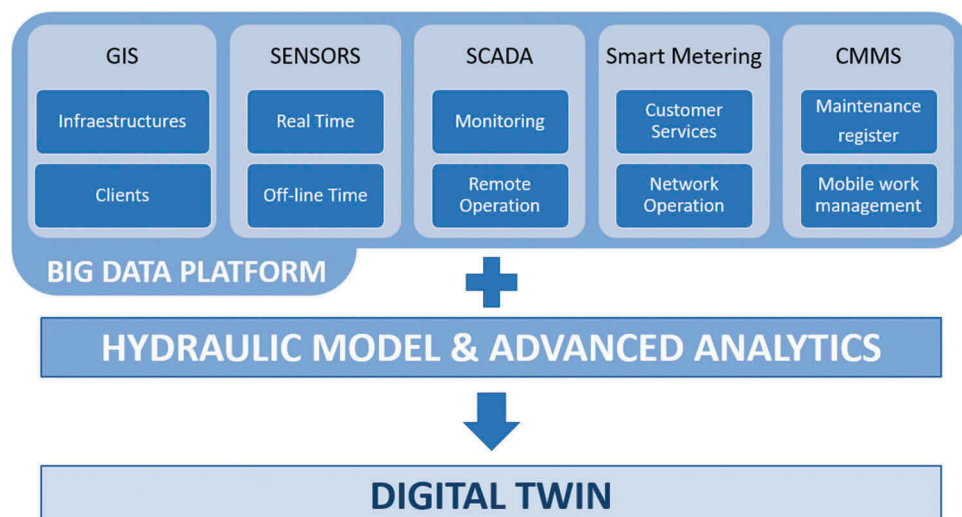


Figure 1. WDS Digital Twin structure.

adjustments and learning, which enriches the hydraulic model with day-to-day knowledge, making use of the machine learning or deep learning techniques supported by large amount of field data stored in big-data platforms.

According to the previous requirements, the authors have designed a strategy to develop and maintain updated a DT of the water distribution network, integrating the hydraulic model with all the information sources (Bartolín, Martínez, and Cortés 2006)

A new application GO2HydNet, whose structure is sketched in Figure 2, was developed by the authors with the aim to build automatically from scratch, and by querying the different information sources, an Epanet-based detailed model for a selected area and a certain time period. The detailed model can include all the pipes, operating elements and auxiliary elements that affect the water flows, such as tanks, regulating valves, pumps, shut-off valves, hydrants, drains, air release valves, service connections, sensors, etc.

The application rebuilds the topological relations of all read elements from their geographical coordinates. Pumps and regulating valves are imported as small linear elements with their own output and input nodes. Furthermore, the application contains tools to detect any error, both regarding the elements' connectivity and their properties declaration. In order to solve those errors while keeping the process alive, GO2HydNet can be fed by auxiliary information, which contain instructions to add, modify or remove any element retrieved from the GIS database.

The application has also additional tools focused on the retrieved data management. In order to avoid redundancy of information, it can optionally remove aligned vertices and merge consecutive pipe segments if their diameter, material and age coincide. Moreover, in case confluences without pipe splitting are detected, the application can create T-shaped connections. GO2HydNet is also able to detect isolated areas and to remove them if neglectable due to their size.

Finally, additional properties not defined in the GIS can be estimated and assigned to the model elements. For example: the pipes roughness is calculated by means of their material

and age; the nodes elevation are obtained by interpolation of the Digital Terrain Model (DTM), freely available at the Spanish National Geographic Institute; the pipes status can be established depending on the status of the superposed manual valves; the flow and pressure meters can be linked to their corresponding pipe or node respectively; and the District Metered Areas (DMAs) can be delimited by exploring the network and identifying the DMA boundaries where flowmeters and closed manual valves exist.

Once the physical model is complete, GO2HydNet implements the processes to load the model by using the measured field data of a given time period (Bou, Martínez, and Conejos 2006). First, the nodal base demands must be assigned. In case the consumptions are remotely available thanks to the smart metering, the values can be directly applied to the service connections (if considered in the model, otherwise they are applied to the nearest nodes), defining a different pattern to each node. The variability of demands over time can also be simplified, establishing a single demand pattern for each DMA. In such case, the pattern is obtained from the DMA's inlet flows.

Second, the information stored at the CMMS is retrieved to check if there were isolated zones due to maintenance operations during the queried period. In such case, the corresponding pipes statuses of the model are accordingly modified, and the demands of the isolated area are cancelled during the operation, to reproduce the real behaviour of the network.

Finally, the application retrieves the field data from the SCADA system corresponding to the queried time period. These data permits, on the one hand, to adjust the model demands with the aim to reproduce the flow supplied by the tanks and reservoirs and, on the other hand, it allows to determine the control rules required to replicate the behaviour of all regulating elements.

This detailed model built by query represents the network behaviour for a certain time period in the past (which can span to one or several days), but also can be connected with the SCADA to make live simulations. Hence, it can be used as an assistant to make previously tested real-time decisions. However, sometimes, depending on the use, connecting with

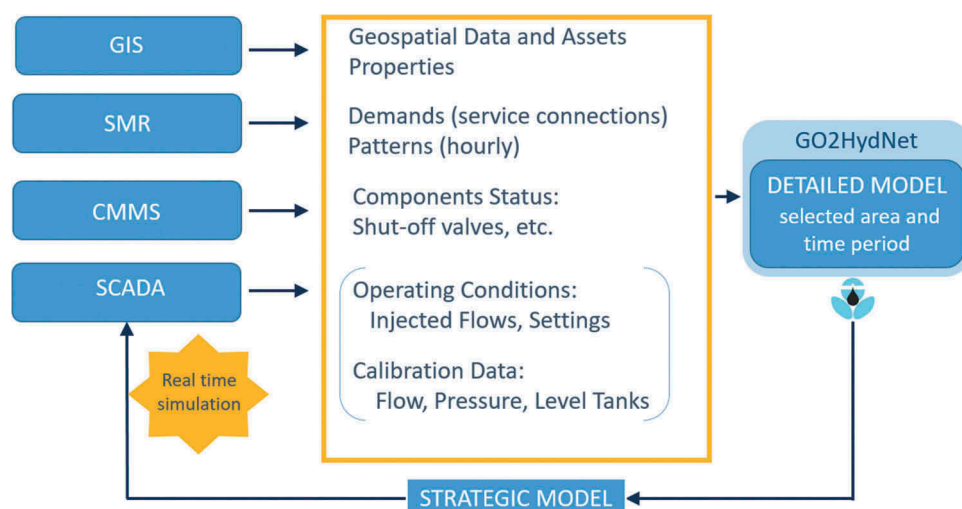


Figure 2. Diagram of the proposed strategy to integrate the hydraulic model with the different information sources, implemented in GO2HydNet application.

the SCADA a strategic model containing only the main elements is more useful to have a general view of the system behaviour. In that case the detailed model can be simplified automatically to obtain a strategic model, maintaining both of them connected.

Either with the strategic or the detailed model, the network behaviour can be simulated any time: in real time, at any past time or near future (next 24 hours) considering the real network conditions, which is an essential part of a DT.

It is important to highlight that reaching this goal is not an easy task, and several challenges were overcome for the case study. Some of them are listed below, but similar challenges will probably be faced for any other particular cases.

- (1) It was necessary to develop algorithms to fit GIS information to the model requirements:
 - GIS and model topology are different, for example, valves are usually represented as a point in the GIS while in the model are links.
 - GIS information is sometimes wrong, for example, it may have inconsistent properties, failures on the network topology, etc. Hence, it was necessary to develop automatic error detection and correction tools to feed the model.
 - Finally, GIS may not contain all the necessary information, e.g. pump curves, pipe roughness, etc., so a system to store all the necessary extra information was designed.
- (2) Algorithms to calculate users' demands were developed. This is necessary because the consumption is registered with different types of meters, and therefore in different time frequency. For example, some of them are smart meters that register the consumption every hour, others every day or even every week, some others are analog meters and the consumption is registered every certain months.
- (3) On the other hand, developing algorithms to filter and correct the raw information provided by the SCADA was necessary, because faulty data leads to faulty simulations.
- (4) Moreover, the elements considered by Epanet are limited. Three methodologies could be used to include the new elements considered in the detailed model. On the one hand, a new single component could be represented by two or more combined components available in EPANET. On the other, iterative methods could be implemented externally to fit the real behaviour of some components, by properly using the functions provided in the EPANET programmers' Toolkit. At last, the source code of EPANET could even be modified to account for new elements.

Finally, it is important to justify the solution adopted to keep the model updated. As described above, the model is built each time from scratch instead of synchronizing it with sources, i.e. updating in the model just the changes occurred at sources.

The low time required to build the model from scratch (just a few minutes for a complex network of up to 100,000 nodes) allows to proceed in this way and, thanks to that, facing the

complexity of synchronization processes is avoided, while the coherence of the model as a whole is always guaranteed.

On the other hand, in order to reproduce again any past situation, a historical database of the changes made on the source data is kept active. These changes refer to GIS information, filtering of field data, restitution of registered consumption, etc. Maintaining a history of changes also avoids having to rewrite them for the next scenarios, and an assistant will facilitate the work of updating such database.

Results

According to the previous requirements, Global Omnium, GoAigua and the Universitat Politècnica de València (UPV) defined a strategy to develop and maintain a DT of the water distribution network of Valencia and its Metropolitan Area, which approximately supplies water to 1.6 million inhabitants. The whole network has a total of 113,000 pipes, 8 reservoirs, 28 tanks, 47 pumps, 259 regulating valves, 48,500 manual valves, 4,600 hydrants, 118,000 service connections, 97 flowmeters, 470 pressure gauges, etc.

In a first phase, the application GO2HydNet, built from scratch, was used to obtain from the GIS system a detailed model of the whole network or a selected area, which can include all the pipes, operating elements and auxiliary elements that could affect the water flows. The exploration algorithms to build the topology and spatial relations were developed in order to achieve the highest possible computational efficiency. Thereby, the process of building the detailed model of Valencia for 24 hours, including service connections with their corresponding consumption pattern each one, to reach a complete model of 325,000 nodes (about 100,000 from the network topology and two additional nodes for each service connection), takes about 1 minute of computation time on a personal computer with an i7-3.2 GHz processor and 32 Mb RAM, employing one single processor. The process included data retrieving, spatial processing and writing the Epanet input file. The time required for data pre-processing was not included in the analysis, neither that needed by Epanet to open the input file, which is longer than the time above.

The detailed model automatically built thanks to the GO2HydNet application can reproduce the network behaviour for any past time spanning from one to several days, and it considers all the maintenance operations occurred during the simulation period (i.e. manual valves closure), as they affect the consumptions, flows and pressures surrounding the working area. Finally, the field data (up to 20,000 daily records) are processed to set the nodal demands and the control rules of the model in order to reproduce the real behaviour of the network up to the minimum detail allowed by the data sources, using the Epanet engine. [Figure 3](#) shows the extension of the whole hydraulic model build by request, and a detailed view of the service connections on the city centre.

However, this model is too big to be used in real time so, one of the future actions will be the development of an algorithm able to automatically simplify that detailed model in order to reduce it to a more operational strategic model of 10,000 pipes. For now, the Valencia's DT works with a base strategic model that nowadays is not linked to the detailed model, and so, must

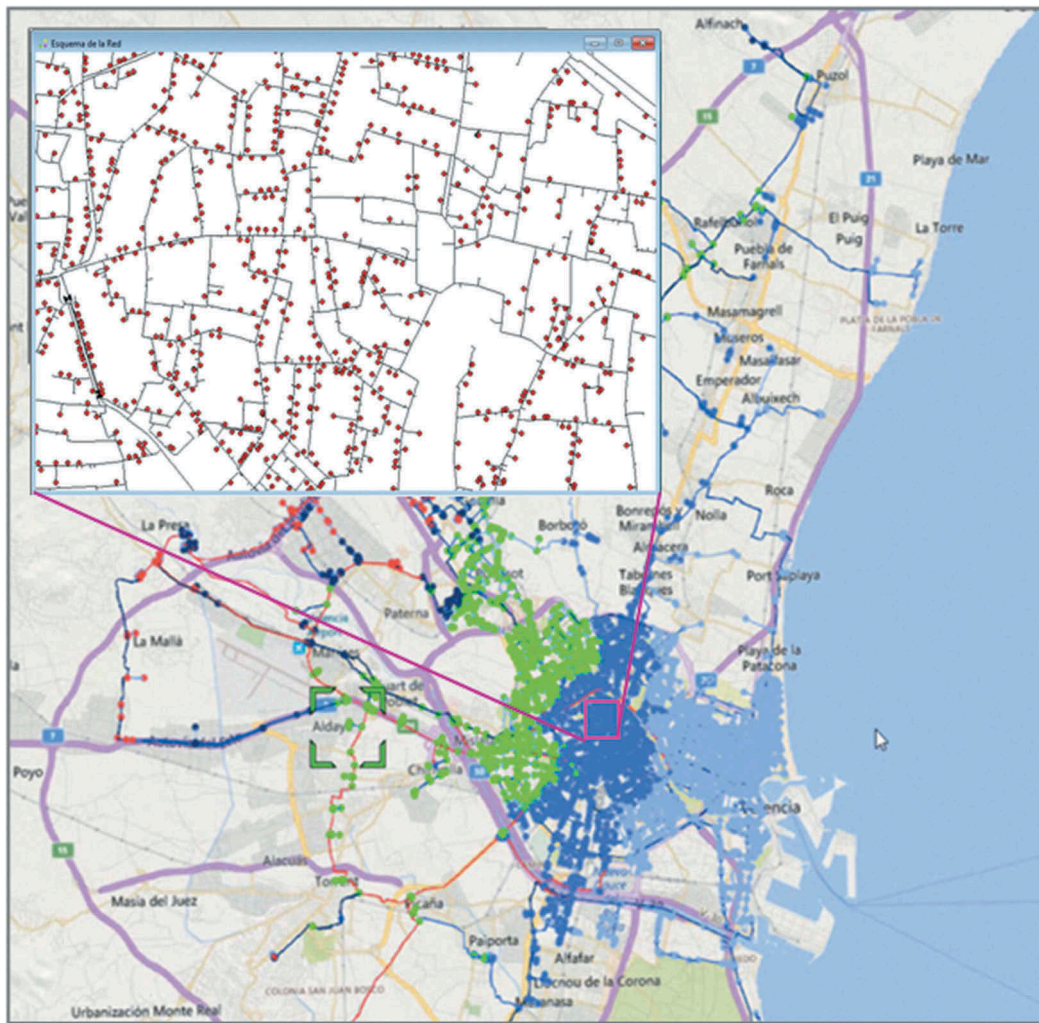


Figure 3. Building the Valencia's detailed model of 325,000 nodes, including service connections, by query. The zoomed area shows a particular zone of the city centre where red dots stand for demands in service connections.

be updated every certain time. This DT is coupled with the field data registered by the SCADA system in order to adjust the demands, pressures, levels and status of the regulation elements at every time instant, in order to get an operating twin that is updated every 1 minute. With the adjusted strategic model, the operators can simulate the real-time decisions, analyse the system behaviour at any past time, or program the actions to be carried out in the near future, usually in the next 24 hours.

The strategic model is updated periodically by using the GIS information, the information provided by the smart meters and, if necessary, the CMMS operations (long-term operations). With the proposed strategy, the simplified model keeps its connection with the detailed model, so every operation can be performed with either one or another model, in such a way that both together constitute the DT currently managed by Global Omnium (Figure 4).

Regarding the visualization capabilities of the Valencia DT, various environments were developed. On the one hand, general synoptic views were designed with the aim to provide a DT interface oriented to the network operators. The synoptic views allow simultaneous observation and

comparison of the raw measured variables, filtered variables and simulated ones. Moreover, the information is organized by different colour codes. Each remote station is represented by its own synoptic view, where the user can perform any simulation, in a similar way than a real SCADA, and analyse its impact over the network. The synoptic views also contain additional information, such as water balances, evolution charts or piezometric profiles.

Likewise, the simulated values can be displayed over a geographic background. This view was thought to provide a DT interface for planning tasks. It allows monitoring in real time the hydraulic variables of the whole network. The main advantage of the geographical view is the possibility to approximate the details of the hydraulic state of the different DMAs and supplied municipalities, without the need to install multiple sensors along the network.

The DT of Valencia WDN supports the daily operation of the system and serves multiple uses.

For planning, the DT is mainly used in the development of master plans, for asset management and in the development of contingency planning and resiliency management, where it allows to:

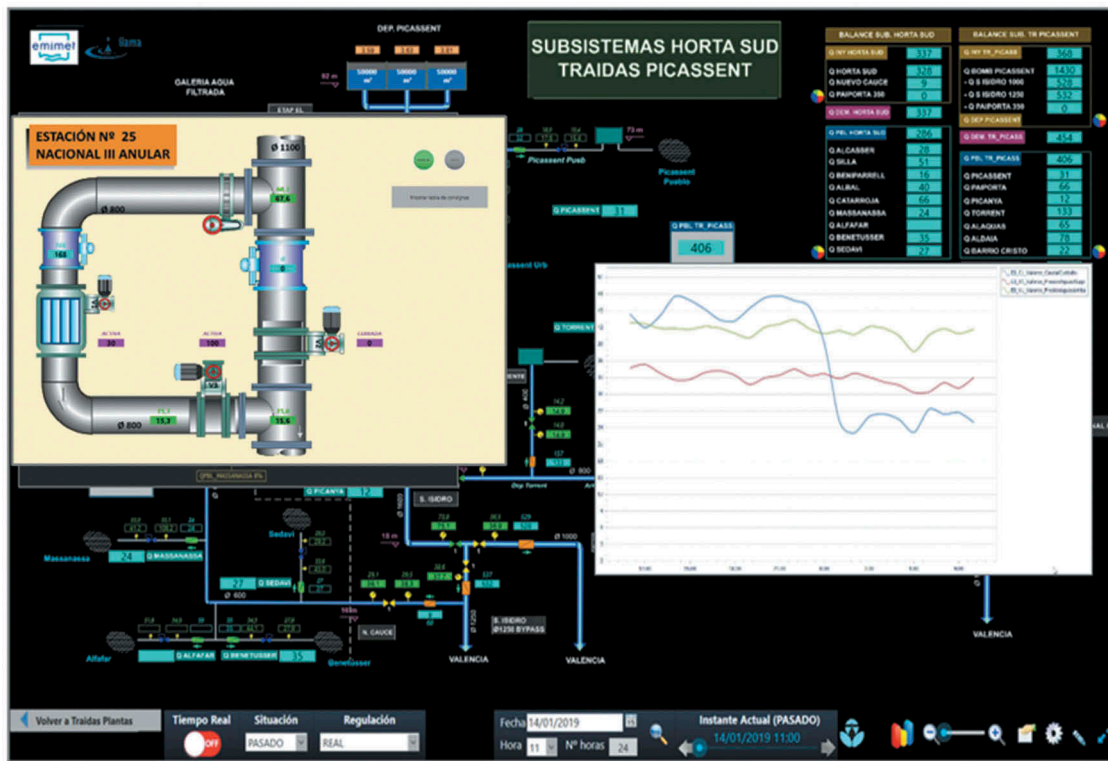


Figure 4. The DT of Valencia WDN connected in real time with field data. By means of double-text boxes real-time data are compared with simulated ones. From the synoptic views of control stations, the proposed actions can be simulated before carrying them out.

- Simulate the network behaviour under long-term demand forecast. This helps the design of new network extensions due to urban expansion. Also, it permits to validate the existing network capacity to assume new supplies.
- Design the replacement plan of existing insufficient or aging infrastructures.
- Design the operation protocols in the process of incorporating new network extensions.
- Define future operating guidelines and operation rules with the projected network configuration.
- Simulate hypothetical emergency scenarios, which allows to determine the actions that must be taken in an emergency and the infrastructures needed

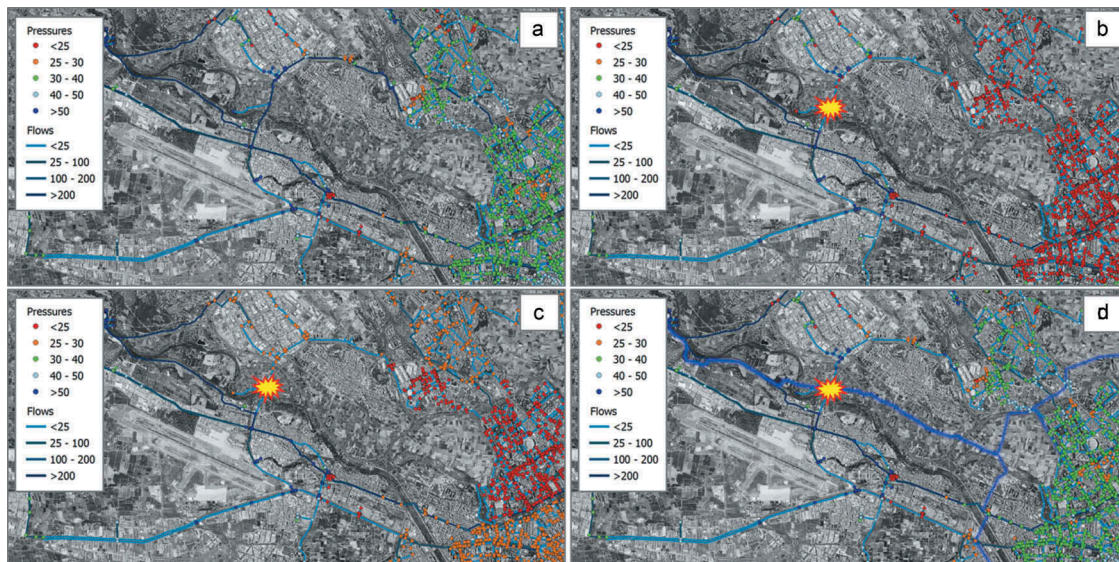


Figure 5. Example of the exploitation of the Global Omnium's DT as a decision support system for simulating a hypothetical emergency scenario. (a) Results under normal conditions. (b) Results under a failure scenario (pipe burst). (c) Results of the best operation strategy to mitigate the effects of the burst. (d) Results with the new infrastructure needed to amend the effects of the burst.

for minimizing the risks and their consequences (Figure 5).

For operation and maintenance, the DT serves as a decision support system for many applications such as:

- To achieve an extended and deep knowledge of the network behaviour, by simulating in real time what is happening every moment at any point (Figure 6).
- To predict the system behaviour in the near future, which commonly considers 24 hours (Figure 7), and to analyse what happened at any past time.
- For arranging coming works to be done in the network, so the best day and schedule can be chosen, and the necessary operation protocols defined (Figure 8).
- For training operators by making them taking decisions under both normal and emergency conditions.
- Acting as a decision support system in real time under unforeseen emergency conditions (Figure 9), as it helps with the development and validation of emergency protocols by live testing or simulating different actions before making decisions.
- To optimize the network operation by minimizing the energy consumption, preserving quality and hydraulic service levels, minimizing leaks, etc.
- To detect anomalies, by comparing the simulated with the observed values, and by identifying outliers. The anomalies identification has been key to prevent downtimes due to elements malfunction. It has helped the utility to be more proactive than reactive.

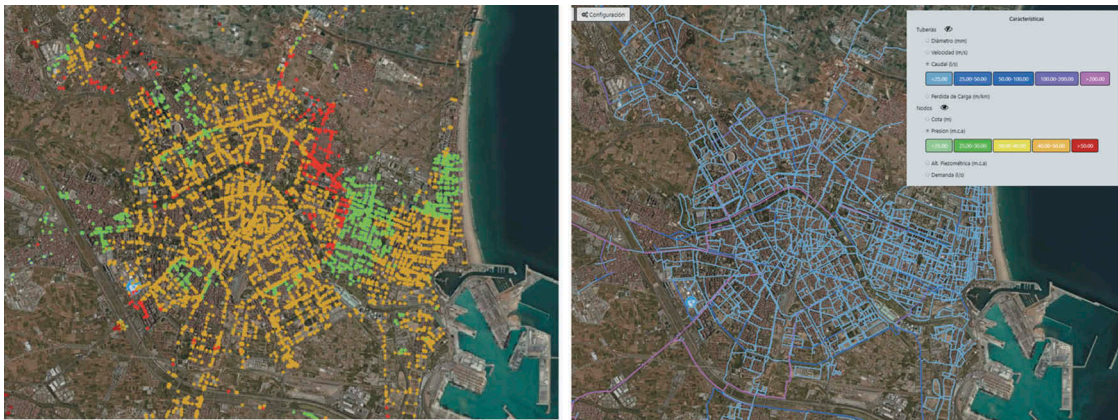


Figure 6. With only 400 measurements (pressures, flows and levels), it is possible to know in real time all flows and pressures of a 10.000 node strategic model.

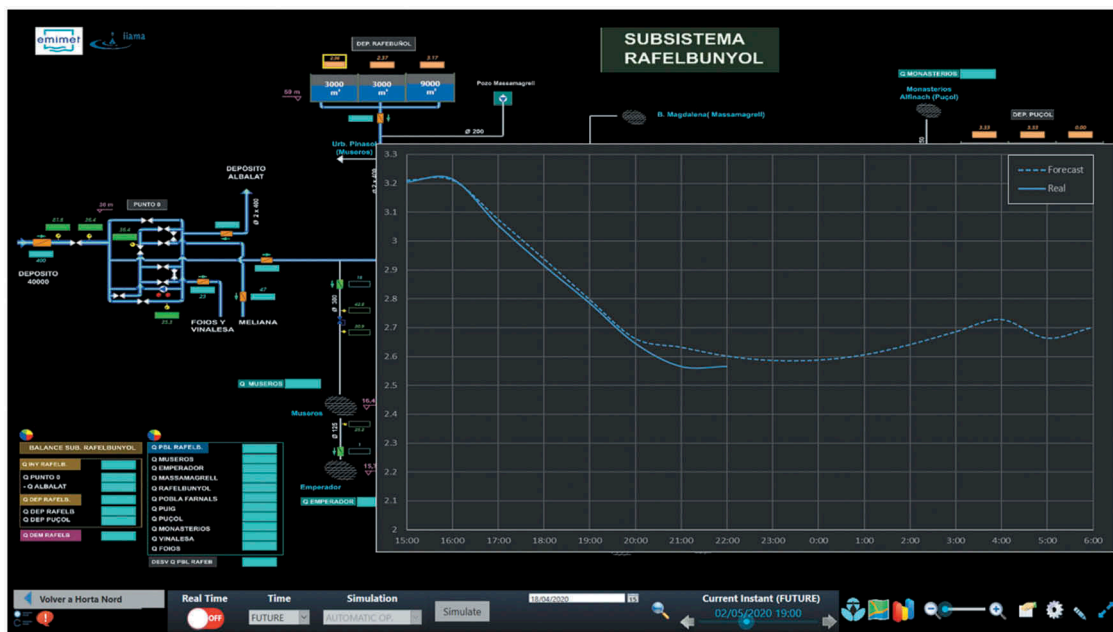


Figure 7. Forecasted tank level during a repair work in the network (dashed line) and real tank level (continuous line). The forecast was performed for a longer period in case the works took more time than expected.

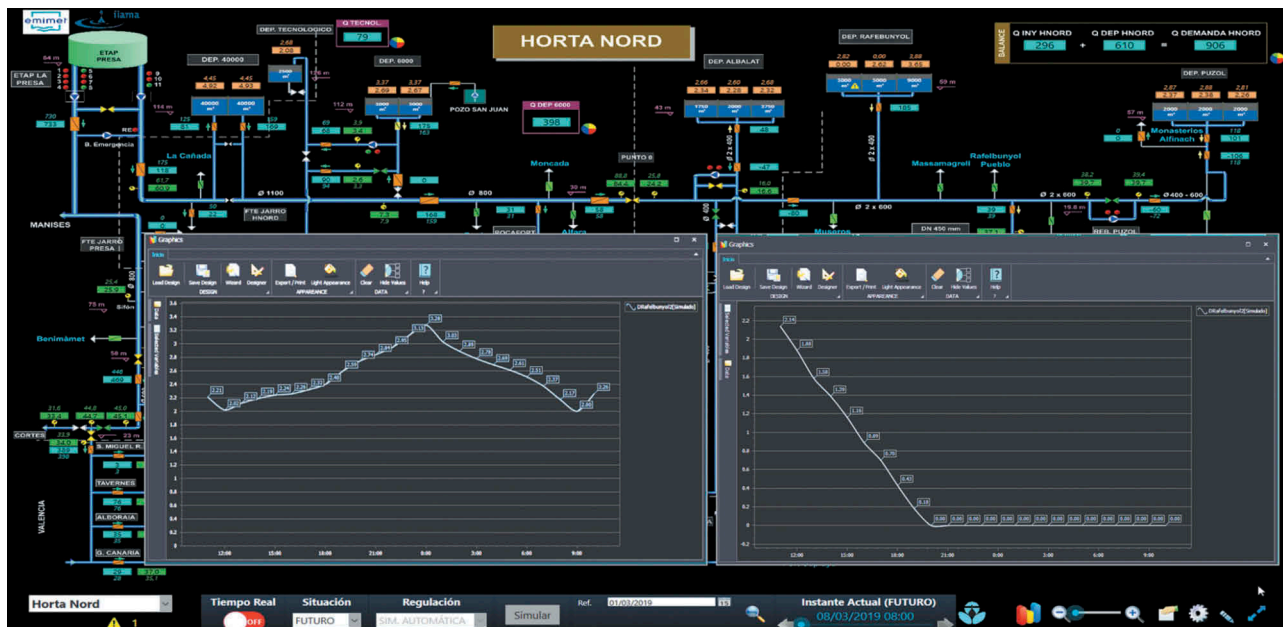


Figure 8. Forecasted tank level evolution for a near future in normal conditions (left) and under a network operation that implies the tank emptying, e.g. tank cleaning or programmed maintenance stoppage at a treatment plant (right).

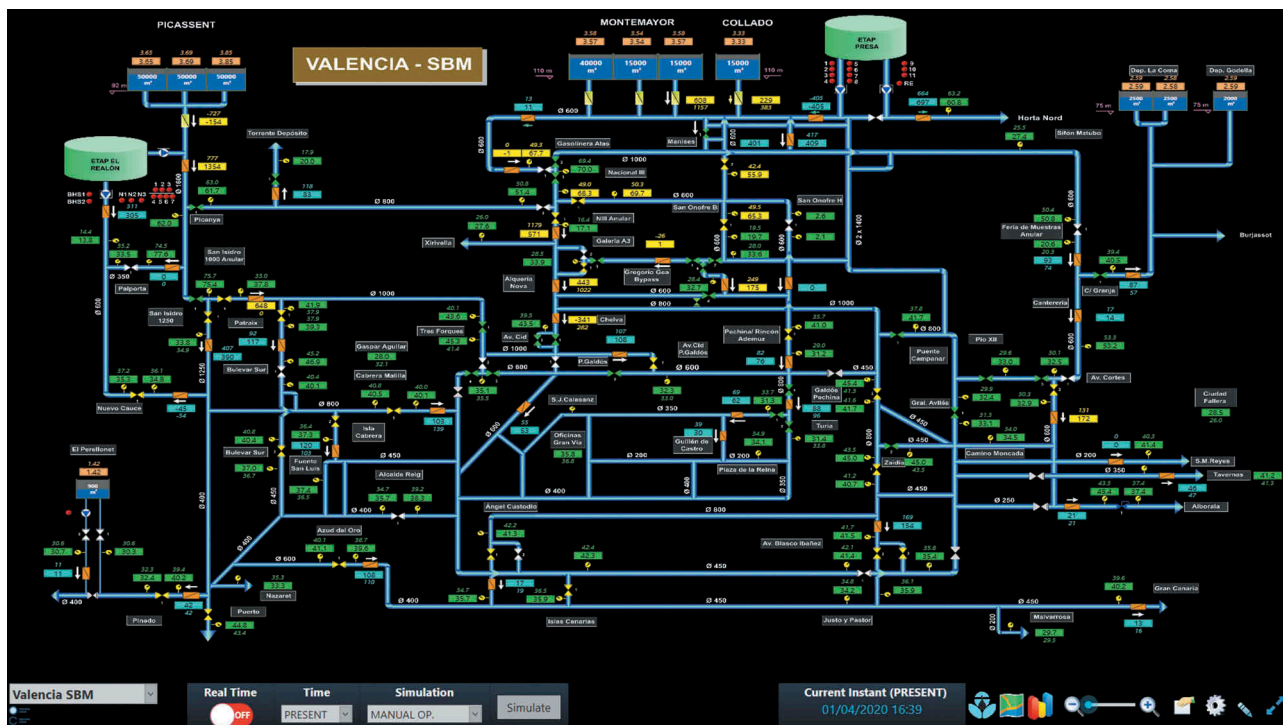


Figure 9. Simulation of a pipe failure in real time. The yellow boxes represent the area of the network most affected by the emergency (pressure and flow variations above 20 %).

Conclusions

Digital Twins have reached the WDS area to stay. The development and live maintenance of a DT of a water distribution network is the further objective pursuit by many water utilities, but it is not an easy task. Nevertheless, the data and tools available nowadays make it possible. This paper presents the efforts made by Global Omnium, GoAigua and the UPV to achieve this goal, becoming one of the first utilities to have

a DT connected to the main data sources of the company. The great benefits ensure that it will begin to be habitual in the most advanced smart cities.

The application GO2HydNet is capable of building a detailed hydraulic model from the information stored at the utility big data platform, which gathers all the information provided by GIS, AMR, CMMS and field sensors. The algorithms were developed with the aim to achieve the highest possible efficiency.

Thanks to that, GO2HydNet is able to build a detailed model with 325,000 nodes in barely a minute.

The processes to load the hydraulic model from field data, as well as their pre-processing and filtering, were successfully designed and implemented. It allows the DT to reproduce any past scenario, as well as to simulate in real time conditions and obtain the forecasted evolution of the near future.

However, there are still some remaining tasks to develop. So far, the detailed model and the strategic simplified model are not yet connected. Further efforts towards an automatic process to connect both models are needed. Moreover, the water quality model has not been developed yet. Due to the growing concern of WUs about the control and improvement of water quality issues, it is essential to take advantage of the potential usefulness of the DT quality model.

Finally, more tools to help managers in the decision-making process and to optimise the network operations are still pending to be integrated.

Disclosure statement

The authors declare non conflicts of interest.

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